Bank Reviews Sentiment Analysis (DFS182-SS2) – Bernette Lesejane

Quick Start Guide

Package contents (**5 files** all in all)):

- Python notebook (bank_reviews_sentiment.ipynb)
- Dataset **(bank_reviews3.csv)**
- Python requirements **(requirements.txt)**
- Peer_Reviewer_Guide **(MS Word file)**
- **SS1 project (startup) –** PDF for reference purposes

1. Extract the ZIP file to a folder on your computer. **Keep all included files together.**

2. Install Python 3.11 or later if it is not already installed. Python 3.14 was used to verify the notebook in this project environment.

3. Open MS Command Prompt (CMD) in the **extracted folder** and create a **virtual environment**:

```
py -m venv .venv
```

4. Activate the environment and install dependencies:

```
.\.venv\Scripts\Activate.ps1

python -m pip install --upgrade pip

python -m pip install -r requirements.txt
```

5. Open **bank_reviews_sentiment.ipynb** in VS Code with the Jupyter extension, select the .venv Python kernel, and choose Run All. Alternatively, run: python -m jupyter notebook bank_reviews_sentiment.ipynb

6. On the first run, allow internet access so NLTK can download its stopwords and WordNet language resources. The analysis runs on CPU; no GPU is required.

7. Review the dataset summary, sentiment classes, train/test split, TF-IDF features, Naive Bayes and Random Forest evaluation, and the generated charts and word clouds.

Peer-review note: sentiment labels are imbalanced. Consider the per-class precision, recall and F1 scores as well as overall accuracy.

Troubleshooting: If the CSV **(bank_reviews3 dataset)** cannot be found, confirm it is beside the notebook and that the notebook is running from the extracted folder. If an import fails, confirm package installation completed and that the .venv kernel is selected.

NOTE: Should you experience any issues, please feel free to contact me on the contacts below:

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Kind Regards,

Bernette